PHASE 2 · DATA SOURCING · CORRECTED 6 SEPTEMBER 2026

Phase 2 datasets

Which second exploitation catalogue to use, and the four reasons the obvious choice does not work as shipped. Supersedes the 5 September version, which contained an error that would have produced a silent failure in the estimator.

Companion to **LEV-calibration-Phase1.pdf** · machine-readable version in **DATASETS.md** · prompt in **RESEARCH-PROMPT.md**

VulnCheck KEV is a superset of CISA KEV, not an independent list.

Their documentation: “encompasses all vulnerabilities listed in the CISA KEV catalog, plus approximately 80% more.” Lincoln–Petersen on the pair as downloaded gives `m = |L1|` by construction and collapses to `|L2|`. The arithmetic runs and returns a plausible number — the worst kind of bug. **It is still the right source**, but only after the filter in §1.

CORRECTION 1

Constructing an independent list from VulnCheck

The schema exposes provenance per report, so a CISA-free list `L2'` can be built rather than assumed. `cisa_date_added` is the marker of contamination, not a convenient join.

1	Drop `cisa_date_added` entirely.
2	Filter `vulncheck_reported_exploitation[]` to reports whose URL host is not `cisa.gov` **and not** `shadowserver.org`. Found on first use: Shadowserver's dashboard is VulnCheck’s largest source, and all 969 Shadowserver honeypot CVEs appear in it, so leaving it in makes VulnCheck contain the SENSOR list.
3	Keep a CVE only if a report survives step 2, **or** `reported_exploited_by_vulncheck_canaries == true`.
4	First-observation date = `min(date_added)` over surviving reports.

> The canaries flag is VulnCheck’s own honeypot telemetry, structurally independent of CISA. **That subset alone is a defensible L2** if the URL filter draws reviewer objections. Document the filter in the paper — it is a methodological contribution in its own right, and no prior work does it.

CORRECTION 2

Four sources measure the wrong quantity

Exploit-DB, Metasploit, Nuclei and AttackerKB record exploit *availability*, or analyst assessment of it — not exploitation in the wild. A CVE with a Metasploit module and no observed attack traffic is a capture in that list and a non-capture in CISA KEV, so including it estimates “CVEs with public tooling”: a different and much larger population.

> **Exclude all four as capture lists. Keep them as covariates.** A CVE with a public PoC is more likely to be captured by *every* list, which is exactly the dependence the model has to absorb. This is an estimand objection, not an access one.

CORRECTION 3

Circularity with EPSS — and it reaches Phase 1

EPSS's exploitation labels come from Fortinet, AlienVault, Shadowserver and GreyNoise (Jacobs et al., arXiv:2302.14172). LEV is computed *from* EPSS, so validating LEV against GreyNoise or Shadowserver is partly circular.

> **Worse, and this hits work already done.** CISA KEV is an input *feature* to EPSS v3 — "Public vulnerability lists (3 features): CISA KEV, Google Project Zero, Trend Micro ZDI." Phase 1 therefore compared a predictor that has the outcome baked into it. Measured in `analysis/leakage.py` with two designs that cannot select on the outcome:

TEST	RESULT
EPSS across the listing date, net of non-KEV controls with the same starting score (1,168 CVEs; pairs straddling a retraining excluded)	median ×1.00; upper quartile ≥ ×2.85
LEV before CISA listed them (220 KEV members with a pre-listing history, median 2.8 months)	15% at LEV ≥ 0.5; median 0.005, against 1.0 afterwards

The leak is real for a sizeable minority of listed CVEs, not all of them. The larger finding is that **high LEV among KEV members is mostly a consequence of listing, not a forecast of it** — which bears directly on CSWP 41's proposal to use LEV to find exploited CVEs missing from KEV.

> **Withdrawn.** An earlier version of this page reported over-confidence rising from 11.9× to 32.7× on a "leak-free" cohort, and a compressed discrimination range there. That cohort was built by deleting KEV members only, keeping every non-member, so the observed rate fell by construction. Both results are withdrawn.

CORRECTION 4

Independence is violated everywhere — model it, do not assume it

Every catalogue in this space reads the same vendor advisories and the same security blogs. Naive two-source Lincoln–Petersen assumes independent capture and will be rejected in review.

`Log-linear MSE`	Interaction terms across ≥ 3 lists, so positive list dependence is a fitted parameter rather than an assumption.
`Stratification`	By vendor tier and disclosure year. Phase 1 already measured capture probability varying 14× by vendor; pooling gives the classic downward-biased estimate.
`Chao lower bound`	Robustness check — consistent under unmodelled heterogeneity, and gives a defensible floor.

A third list is what makes the log-linear approach possible at all.

RANKING

Which sources, in order

#	SOURCE	VERDICT
1	`VulnCheck KEV Community`	**Best.** The only source with per-report provenance and dates, so independence can be *constructed* rather than asserted. Free, one API key, single bulk download.
2	`CIRCL Vulnerability-Lookup`	**Best third list — now used.** No auth. Checked 2026-09-28: 969 CVEs from Shadowserver's honeypot feed and 2,396 from Telegram and similar channels, each sighting with its source. **Its aggregated KEV catalogue contains all 1,676 CISA KEV CVEs** — classify by source and exclude that part, or capture–recapture collapses.
3	`ENISA EUVD`	**Checked and not used.** `/api/search?exploited=true` returns the same 1,740 consolidated entries as the KEV dump, so it is equally contaminated. EU KEV proper is 66 CVEs; 55 are on CISA KEV and 88% of those carry CISA's own date, so it is derivative. *(Corrected: previously recommended the exploited-flag search as the clean route.)*
4	`GreyNoise`	Strong data and genuinely independent of CISA, but EPSS-circular per correction 3, and access is reviewed rather than instant. Apply now regardless — it is the long pole.
5	`Project Zero 0day ITW`	Small N, browser and mobile skewed, and itself an EPSS feature. Not a second list; a high-precision stratum at best.
6	`Shadowserver (direct)`	Effectively unobtainable — reports are scoped to IP space you can prove you own. Route via CIRCL instead.
—	`Exploit-DB, Metasploit Nuclei, AttackerKB`	Excluded on **estimand** grounds (correction 2), not access grounds.

> **If only one: VulnCheck KEV Community.** It is the only candidate where the schema itself lets you demonstrate independence to a reviewer, rather than argue for it. Everything else requires an assumption you cannot support with a field name.

ACCESS

Where to get them

NO AUTH Fetch today

CISA KEV	`cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json`
CISA KEV history	`github.com/cisagov/kev-data` · `github.com/hrbrmstr/cisa-known-exploited-vulns` (MIT, daily snapshots)
EPSS history	`github.com/empiricalsec/epss_scores` — ~2.5 GB, no licence declared
ENISA EUVD	`euvdservices.enisa.europa.eu/api/search?exploited=true` · id map at `/api/dump/cve-euvd-mapping`
CIRCL sightings	`vulnerability.circl.lu/api/sighting/?vuln_id=CVE-XXXX-NNNNN`

FREE KEY Minutes to obtain

VulnCheck KEV	`api.vulncheck.com/v3/backup/vulncheck-kev` with `Authorization: Bearer <token>`. Take the bulk ZIP, not per-CVE calls. 1,000 req/min.
GreyNoise Community	`api.greynoise.io/v3/community/{ip}` — signup **rejects gmail**, use the institutional address

REVIEWED Start first — human turnaround

GreyNoise Research	`info.greynoise.io/community/research-program-request` — full Enterprise API, free for research. **The long pole; apply now.**
Shadowserver	`transform.shadowserver.org/api2/` — normally granted only for IP space you can prove you own. Treat as unavailable.

PRIOR WORK

Nothing found on either question

Capture–recapture on catalogue completeness	**None found.** Nearest adjacent work is descriptive overlap comparison, not estimation. This is the strongest novelty claim available.
LEV evaluation since May 2025	**None found peer-reviewed.** Cite and differentiate: a reference implementation (RiskBasedPrioritization/LEV); a theoretical objection that CSWP 41 misreads EPSS as past rather than next-30-days exploitation; Parla (arXiv:2411.02618); and arXiv:2506.01220, which reports 48 exploited CVEs caught by neither KEV nor EPSS individually — the closest existing result. NIST itself states it is seeking partners with real-world datasets to evaluate LEV; cite as motivation.

> **This was a targeted search, not a systematic one.** Before submission, sweep arXiv cs.CR, WEIS, eCrime, ACM CCS and IEEE S&P from June 2025 onward for LEV specifically. One missed prior evaluation costs the novelty claim.

> **Licensing is still unresolved.** VulnCheck's Attribution Notice covers use “in your own production, solution, or service” and is *silent* on peer-reviewed publication of derived data and on archiving raw data. Email `community@vulncheck.com` and get it in writing. The EPSS history repository declares no licence at all.

ACTIONS

Do this week

1. Register VulnCheck Community, pull `/v3/backup/vulncheck-kev`, and build `L2'` per the §1 filter. That unblocks the whole study.
2. Apply to the GreyNoise Research Community. Reviewed by a human, so it is the long pole.
3. Email `community@vulncheck.com` on publication and archival rights, in writing.
4. Run the systematic prior-art sweep before committing to the novelty claim.

Corrected 6 September 2026. Sources: docs.vulncheck.com/community/vulncheck-kev · arxiv.org/abs/2302.14172 (EPSS v3 features and labels) · vulnerability.circl.lu · euvd.enisa.europa.eu/apidoc · info.greynoise.io/community/research-program-request · github.com/cisagov/kev-data · github.com/empiricalsec/epss_scores. Leakage figures from analysis/leakage.py; see README threats 41–44.